Vinay R Jumani

+91 75500 56701 | vinayrjumani@gmail.com | LinkedIn | GitHub | Portfolio

Education

Birla Institute of Technology and Science (BITS Pilani), Goa Campus *Goa, India*

B.E. (Hons.) Electronics and Instrumentation Engineering *Expected May 2028*

Coursework: Deep Learning, Parallel Computing, Operating Systems, Linear Algebra, Probability and Statistics

Technical Skills

Languages: C++, Python, C, CUDA

ML & Systems: PyTorch, Triton, Hugging Face, SGLang, vLLM, llama.cpp, NVML, Prometheus, Linux, Git

Research & Projects

TierMoE: Expert Placement for Batched MoE Decode *Aug – Oct 2026* | ***Under review, MLSys 2027***

- Characterized expert demand under batched decode on **OLMoE-1B-7B, Qwen3-30B-A3B and DeepSeek-V2-Lite** using routing traces and a Belady-oracle simulator. Batching makes each step touch **51% of Qwen3 experts at** B=16, so recency-based caches thrash: llama.cpp's expert cache ran **1.5–3.0× slower** than a layer split.
- Designed a **static per-expert placement** in llama.cpp that keeps frequency-selected hot experts on the GPU and runs cold experts in place on the CPU (avoiding PCIe transfers) with a **sparse cold-expert operator**. Reached up to **1.34× end-to-end speedup** at B=4 on Qwen3-30B-A3B (**1.60**× with a 50% GPU budget) and **1.15**× at B=16.
- Ran a pre-registered evaluation (thread pinning, interleaved repeats) with correctness gates and a GSM8K check; reported where gains reverse (B=32, coarser MoEs).

QIRA-L: Training-Free Token Halting via Closed-Form K/V Transport | *Dr. Amitava Das* *Aug 2026 – Present*

- Developed a training-free intermediate-layer token-halting method that predicts later-layer keys and values with closed-form, layer-specific ridge transport, without modifying the LLM backbone.
- Retained **89.1% of dense LongBench quality** on Qwen2.5-7B-Instruct at **89% of prompt tokens halted** (**+25.7 pts** over best drop baseline).
- Measured **1.50× prefill speedup**; retained **98.3%** of dense quality on Llama-3.1-8B-Instruct at the same budget.

SGLens: Benchmarking & Profiling Suite for SGLang | **sglens** *Jul 2026 – Present*

- Built a benchmarking and profiling suite for **SGLang** on multi-GPU infrastructure: workload models (chat, long-context RAG, shared-prefix reuse, agentic tool-use) and automated **TTFT, TPOT, ITL, percentile latency**.
- Implemented a resumable **96-experiment benchmark matrix** and GPU telemetry pipeline to study throughput, latency and VRAM under different serving workloads.

PressureScope: LLM Serving Telemetry Harness *Aug 2026 – Present*

- Designed an **engine-agnostic harness** correlating application metrics (TTFT, TPOT, throughput), scheduler metrics and GPU/NVML telemetry to characterize near-OOM serving, with deterministic, Poisson and burst workloads.
- Established a reproducible Qwen2.5-0.5B-Instruct baseline on an RTX A6000 with $< 1\%$ **cross-run variance**; validated joint Prometheus/NVML telemetry over **876 correlated samples**.

CUDA GEMM Optimization & LLM Inference | **CUDA MatMul & vLLM Analysis** *Mar – Apr 2026*

- Implemented a **CUDA GEMM optimization pipeline** reaching **92% of cuBLAS performance**.
- Studied the **vLLM codebase and PagedAttention**, FlashAttention, KV-cache management and memory-efficient inference.

Energy-Efficient GPU Autotuning | *Dr. Gargi Alavani Prabhu, BITS Goa* *Aug 2026 – Present*

- Developing an **energy-aware GPU autotuning framework** to find efficient execution configurations across ML/DL workloads and hardware.
- Building **learned performance models** that predict kernel execution metrics to guide configuration search, reducing reliance on exhaustive GPU profiling.

Enterprise Text-to-SQL Interface | *TCS Research, BITS Goa* | **NL-SQL** *Dec 2025 – May 2026*

- Co-architected a Python text-to-SQL pipeline (schema retrieval, ambiguity resolution, SQL generation, execution) for enterprise databases, with Profs. Tanmay Verlekar and Ashwin Srinivasan.

Work Experience

Undergraduate Researcher, Nanyang Technological University *Nov 2025 – Present*

Advisors: Dr. Yuvaraj Rajamanickam (NTU/NIE), Dr. Amalin Prince (BITS Goa) *Singapore*

- Fine-tuned the **LaBraM Transformer** in PyTorch, achieving **96.9% LOSO accuracy** across **73 subjects**.
- Built an attention-based interpretability framework using **Grad-CAM** with feature-level analysis of EEG representations.
- Manuscript under review at a Q1 journal. **EEG-LLM**

Undergraduate Researcher, CSIR-CEERI *May – Jul 2026*

Supervisor: Dr. Madan Kumar Lakshmanan *India*

- Developed and optimized an edge-deployed, non-contact **rPPG vital-sign monitoring system** on the **NVIDIA Jetson Nano** for embedded healthcare applications.
- Integrated automated preprocessing and inference pipelines for **real-time physiological signal estimation**.

Achievements

International Chess Athlete, FIDE 1852 | Vice Captain, BITS Goa Chess Team *2014 – Present*

- Led a 6-member team to **8th globally** at the FIDE Amateur Chess Rapid Team Championship. **Certificate** | **FIDE Profile**